" if you want to manage it Measure it "

IOTAFLOW
SYSTEMS PVT. LTD

UF-1500 Intrusive Ultrasonic Flow Meter

Full Pipe / Partial Full Pipe Applications

www.iotaflow.com

- Digital signal processing, high noise immunity
- DN range from 10 mm up to 5000 mm
- Stable operation in spite of weak signal on aged piping
- reliable metering on fluid parameters alteration
- Easy commissioning
- Bidirectional flow measurement
- Clamp-on and wetted transducers
- No pressure loss in pipeline
- Batching mode to measure the dose
- Multi-beam measuring
- Up to 4 pipes with one electronic unit
- Waterproof transducers
- MID and OIML certified
- ISO 9001:2008 certified

Application:

- Hot and cold process and potable water
- Partial Full Pipe
- Waste water
- Acids
- Alkalis
- Liquid foods

Specification:

Parameter	Wetted PEA	Clamp-on PEA
Nominal diameter of the pipeline, DN	10-5000	50-5000
Accuracy, %, with flowrate range from 1:100: - single-beam sounding - two-beam sounding - four-beam sounding	 ± 2 ± 1 ± 0,5	
Flow velocity range, m/s	0.1-20	
Fluid temperature range, °C	minus 30-160	minus 30-150
Ambient temperature for transmitter, °C	0-50	
Maximum pipeline pressure*, MPa/bar	2.5/25	-
Transmitter / transducers protection	IP54/IP68	
Measurement data logging, number of records: - hourly log - daily log - monthly log - programmable - batching	 1440 60 48 14400 512	
Power supply, VDC	24	
Power consumption, VA	under 12	
Transmitter weight, kg	under 3	
Transmitter dimensions, mm	250x132x90	

*- on request is possible to supply pipeline sections with wetted PEAs designed for up to 25 MPa/250 bar pressure.

DESCRIPTION AND OPERATION

Application

"IOTAFLOW UFM" 1500 series multichannel ultrasonic flowmeter is designed to measure average volumetric flow rate and volume of bidirectional flows of various liquids (hot, cold and waste waters, acids, caustics etc.) in one or several pressure pipelines under various operating conditions including explosion hazard areas Partial Filled Pipe. The flowmeter is suitable for a wide range of applications in power industry, municipal engineering as well as in oil, gas, food and other industries. The flowmeters may be used as part of heat meters, metering systems, automatic process control systems etc.
“IOTAFLOW UFM" 1500 series is suitable for measuring flow rate and volume of liquid food products including alcohol-free drinks (juices, syrups and the like), alcohol drinks, dairy products (milk, yoghurt, kefir, sour cream etc.), mayonnaise, ketchup, solutions of food acids and alkalis etc.

"IOTAFLOW UFM" 1500 series multichannel ultrasonic flowmeter performs the following functions:

- Measuring average volumetric flow rate in either forward or reverse flow directions through 1-4 measuring channels (pipelines)
- **Measurement with 4-Track Insertion Ultrasonic Flow Meter Integrated with a Level Indicator, tailored for accurate flow measurement in partially filled pipes. By utilizing phased probe configurations the system ensures precise flow rate and volume adjustments in real-time.**
- Totalizing volume of forward and reverse flow independently and calculating their algebraic sum with regard to flow direction for each measuring channel
- Determining the current value of flow velocity and flow direction for each channel
- Batching preset liquid volume or batching in the start-stop mode along with determining the amount of batched liquid, batching time and flow rate
- Outputting measurement results to the analog (current), pulse or logical output
- Logging measurement data and configuration settings in the internal nonvolatile memory
- Displaying measurement, diagnostic, configuration and history (logged) data on the built-in indicator, data is transferred via RS-232 or RS-485 interface using dedicated cable, phone cable, Ethernet interface, radio channel or telecommunications channel
- Configuring flowmeter's settings according to on-site and process requirements - Monitoring and indicating alarm conditions (error situations) and faults; recording error and fault types and intervals into the dedicated logs - Providing protection of logged data and configuration settings from unauthorized access.

Specifications

Parameter	Value	Notes
1. Number of measuring channels	From 1 to 4	Optional
2. Nominal diameter of the pipeline, DN	From 50 to 5000	
3. Temperature of the medium, °C	From - 30 to +150	
4. Power supply voltage	DC 24 V	See section 1.2.6
5. Power consumption, W	no more than 12	
6. Mean time to failure, h	75 000	
7. Mean life time, years	12	

Notes.

1. When using overhead electroacoustic transducers (PEA), the minimum nominal diameter is 50 mm.
2. It is possible to use the flow meter for work on pipelines dah from DN up to 10000 mm.
3. The permissible temperature range depends on the design of the application. emitted PEA.
4. Installation of overhead PEA on the pipeline using a specialized set of mounting parts allows you to measure the flow of the controlled liquid with a temperature of up to 350 °C.
5. With the standard design of mortise PEA. Upon request, the flowmeter can be supplied with measuring sections and embedded PEA for pressures up to 25 MPa. 1.2.2. The flow meter provides measurement of average volumetric flow at flow speeds up to 20 m/s in accordance with the formula:

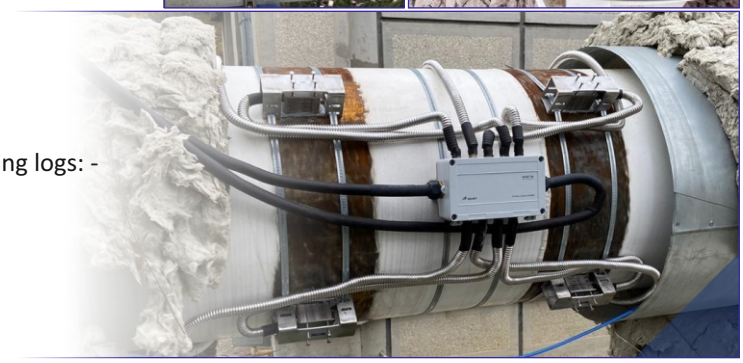
The flowmeter measures average volumetric flow rate up 20 m/s flow velocity on the base of the following formula:

$$Q = 2.83.10^{-3}.v.D^2,$$

Where Q-average volumetric flow rate, m^3/h;
v-flow velocity, m/s;
D-pipeline internal diameter, mm.

Volume is measured at a flow velocity value of no more than 10.6 m/s. This value can be increased up to 20 m/s. Flowmeter's sensitivity to flow velocity is no less than 0.01 m/s.

The flowmeter provides that the results of measurements can be outputted to:
- Universal outputs from 1 to 9 (optional)
- Analog output-2 (optional) - RS-232 (RS-485) interface - 1;
- Ethernet interface-1 (optional).



The flowmeter provides storage of measurement results in the following logs: -
Hourly log-1440 records (hourly records) 60 days
Daily log-60 records (daily records)
Monthly log-48 records (monthly records)

Interval log - up to 14400 records
Batcher log-up to 512 records per a channel
User log-up to 1000 records
Measuring channels error log-up to 512 records per a channel
Failure log-up to 60 records
Mode log-up to 512 records.
When powered off, the flowmeter stores measurement results and configuration settings for no less than 1 year. The flowmeter is powered from a stabilized 22 ... 29 V DC voltage source with pulse level of no more than ± 1.0%. The flowmeter can be powered from mains AC (154-264) V (50±2) Hz via Secondary Power Source ADN-3024 (AND)

Environmental restrictions:

a) Ambient temperature

• CS: ambient temperature from 0 to +50 C (optional frost-resistant model designed for the temperature range of-40 to +65 C is available)
• Clamp-on PEAs: liquid temperature from - 30 to + 150 C

b) Relative humidity

• CS: up to 80% under + 35 C, without moisture condensation
• PEAS: up to 100% under maximum + 40 C, with moisture condensation.

c) Atmospheric pressure from 66.0 to 106.7 kPa;

Protection provided by the enclosure:
• CS: IP54
• PEA: Ip67

Metrological Specifications

Limits of permissible relative error for measurement, indication, logging, storage and transferring of average volumetric flow rate and volume measurement results for any flow direction for flowmeters with primary converters1 (CPs) do not fall outside the following ranges and operation conditions comply with those specified in the present manual.

8 = ±(0,95+)- using single-beam sounding method with a liquid viscosity of ≤ 4 cSt
8 = ±(1,2+)-using single-beam sounding method with a liquid viscosity of > 4 cSt
6=±(0,45+)-using two-beam sounding method
6=±(0,25+)- using four-beam sounding method.

Where 8-limits of permissible relative error, %; v-flow velocity, m/s.
NOTE. 1 "Primary converter" means a pipeline section with PEAs installed on it.

The limits of permissible relative measurement error for measuring time of operation in various modes do not exceed 0.1%.

Outputs

• LCD;
• pulse or logic digital output (up to 8 additional digital outputs on request);
• RS-232/RS-485 serial interface (ModBus); 0-5, 0-20, or 4-20 mA analog output (on request)
• Ethernet (on request

Compound

The composition of the flow meter upon delivery is in accordance with Table

Name and symbols	Qty	Note
1. Secondary measuring transducer	1	Note 1
2. Electroacoustic tansducer	2-8	Note 2
3. Measuring section	0-41	Note 2
4. Installation kit	1	Note 3
5. Calibration certi ate	1	Note 4
6. Operational docume taation - instruction manual, parts I, II - installation instructions		

Notes:

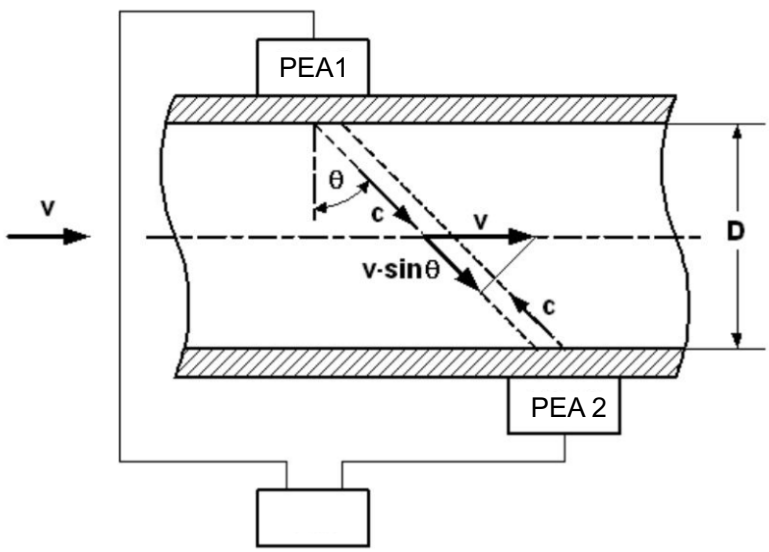
1. The number of measurement channels, as well as the number of service modules external relations departments - upon request.
2. Quantity and type upon order. When the device is delivered with a measuring section, all necessary settings are made at the factory - manufacturer.
3. With standard delivery, the length of the VP-PEA cable is 10 m (2Ѳ10 m per channel). It is possible to supply cables in coils.
4. Operating documentation and order cards for this product and other products manufactured by IOTAFLOW are posted on the website at www.iotaflow.com

There is also a package of programs "Universal viewer", which includes the instrumental program "Monitor IOTAFLOW (URSV-5xx c)" for working with the device via the serial interface RS-232 (RS-485) and the Ether interface -net.

DESIGN AND OPERATION

Principle of operation

The flow meter is a time-pulse type device, the operating principle of which is based on measuring the delay in the passage of ultrasonic signal pulses downstream and against the flow due to signal drift by the controlled liquid.



Electrical probing pulses generated by the VP alternately supplied to converters PEA1 and PEA2 (Fig. 1).

Fig. 1. Diagram of the flowmeter measuring section.

The ultrasonic pulse emitted by one PEA passes through the liquid moving through the pipeline and is perceived by another PEA. When the liquid moves, the ultrasonic wave is carried away, which leads to a change in the propagation time of the USS: along the fluid flow (from PEA1 to PEA2), the passage time decreases, and against the flow (from PEA2 to PEA1) it increases. The difference in the time of passage of the ultrasonic system through the liquid along the flow and against the flow is proportional to the flow velocity v and, therefore, to the volume flow Q.

The digital method of processing the signals received by the PEA ensures stable operation in conditions of interference, and also simplifies the setup of the flow meter during commissioning.

The fluid velocity averaged along the ultrasonic beam is determined divided as:

$$v = \frac{c^2}{2 \cdot n \cdot D \cdot tg\theta} \cdot \left[(T_{П2} - T_{П1}) - dT_0 \right],$$

where c is the speed of propagation of ultrasonic waves in a stationary liquid

D- internal diameter of the pipeline

y-angle between the direction of propagation of the ultrasonic pulse and the plane position perpendicular to the axis of the pipeline

TP1, TP2-the total time of passage of the signal through the measurement channel (VP, communication cables, both PEA, liquid) when the USS propagates along and against the flow, respectively

dTO - difference in signal transit times when the liquid is stationary (flow meter zero offset)

n - coefficient depending on the PEA installation scheme (Fig. 2)
n=1-when installing PEA according to the Z-scheme or in the U-elbow
n = 2-when installing PEA according to the V-scheme.
The flow rate is calculated according to the expression:

$$Q = \frac{\pi \cdot D^2}{4} \cdot v \cdot K_г ,$$

Where Kg = vav/v - hydrodynamic coefficient.

The hydrodynamic coefficient is the ratio of the average fluid flow velocity in the pipeline to the fluid flow velocity v averaged along the ultrasonic beam. It is calculated based on the entered values of the pipeline wall roughness, the viscosity of the controlled liquid, the internal diameter of the pipeline, and the measured flow velocity.

When measuring flow using multi-beam sensing schemes along chords of a controlled flow, the value of the hydrodynamic coefficient is not used. This allows such measurements not to take into account the roughness values of the pipeline walls and the viscosity of the controlled liquid, because The measurement results do not depend on the values of these parameters.

The volume of liquid V for the time interval T is determined in accordance with the formula:

$$V = \int_0^T Q(t) \cdot dt .$$

The change in the speed of propagation of ultrasonic waves in the working fluid, associated with changes in temperature, pressure and/or composition of the liquid, due to the constant length of the acoustic path, is taken into account in the device by determining the half- sum of the time of passage of the ultrasonic waves of the distance between the PEA along and against the flow T:

$$\Sigma T = \frac{T_1 + T_2}{2} .$$

In a two-beam flowmeter, built on the basis of a two-channel meter, two pairs of PEA are installed on one pipeline (PI). In this case, the flow rate for each beam is measured independently, and the measured flow rate in the pipeline is calculated using the formula:

$$Q = \frac{Q_1 + Q_2}{2} ,$$

Where Q1, Q2 - flow rate measured along beam 1 and beam 2, respectively.
In a four-beam flow meter, built on the basis of a four-channel meter, four pairs of sensors are installed on one pipeline along mid-radius chords (four sensors in 2 different sections of the pipeline). In this case, the flow rate along each beam is measured independently, and the measured flow rate in the pipeline is calculated using the formula:

$$Q = W1 \cdot Q1 + W2 \cdot Q2 + W3 \cdot Q3 + W4 \cdot Q4,$$

where W1 = W2 = W3 = W4 = 0.25.

The flow rate is determined when the following condition is met:

$$Q_{отс} \leq Q,$$

Where Q_{OTC} minimum flow rate (lower cutoff), m3/h;

Q-current flow rate, m3/h.

The recommended lower cutoff value corresponds to a flow speed of 0.035 m/s.

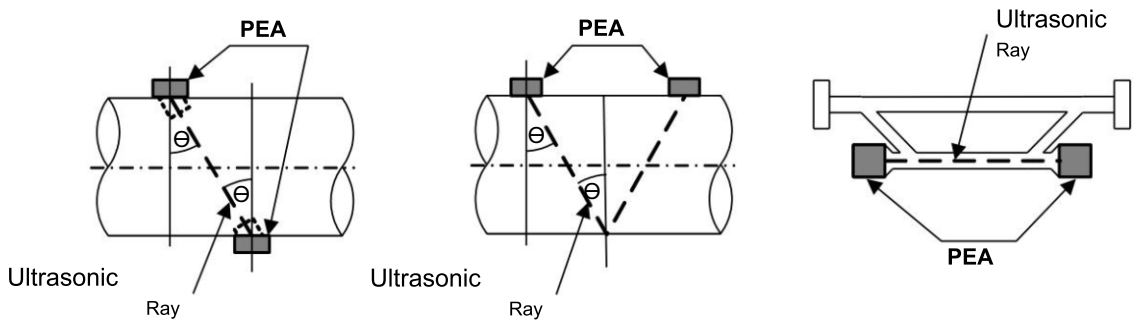
If the condition Q<QOTC is met, then in the flow meter the measured flow rate value is equal to zero, the accumulation of volume stops, the generation of pulses at the universal output stops, and the output current becomes equal to the lower value of the range.

When the condition Q>Qmax is met (where Qmax corresponds to a flow speed of 10.6 m/s), flow measurement continues, but the accumulation and archiving of volume, the issuance of pulses at the universal output stops, and the output current becomes equal to the upper value range.

Overhead PEAs are installed on the outer wall of the control unit without opening the pipeline, mortise PEAs are installed in holes in the walls of the control unit. PEA can be installed on the pipeline according to the following schemes (Fig. 2):

- Z- scheme - PEAs are placed on opposite walls of the IU in a plane running along the axis of the IU (installation "along the diameter"), or parallel to the axis of the control unit (installation "along the chord" - only mortise PEA); in this case, the signal from one PEA to another passes without reflection from the inner surface of the DUT;
- V-scheme - PEAs are installed along one wall of the IU in a plane passing along the axis of the IU; in this case, the signal from one PEA gets to another after reflection from the internal surface of the DUT (at the same value of the angle Ɵ, the USS travels twice as far as with the Z-scheme)

- in a U-shaped PS (U-elbow type) - mortise PEAs are placed at the ends of a straight section of the PS; in this case, the signal propagates along the flow axis.

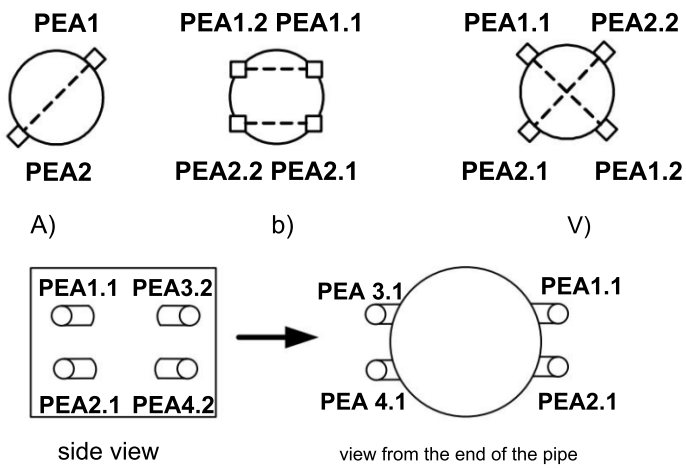


a) According to the Z-scheme b) According to the V-scheme c) In a U-elbow type IU

Fig. 2. Schemes for installing PEA on a pipeline.

The longitudinal angle of installation of mortise PEA in a straight IU (the angle between the axis of the mortise PEA and the plane perpendicular to the axis of the pipeline) can lie in the range from 20Ɵ to 70Ɵ, the recommended value is Ɵ 45Ɵ. At DnƟ2000, the range of permissible values of the longitudinal angle of installation of mortise PEA is determined by the characteristics of the object where the PPE is placed.

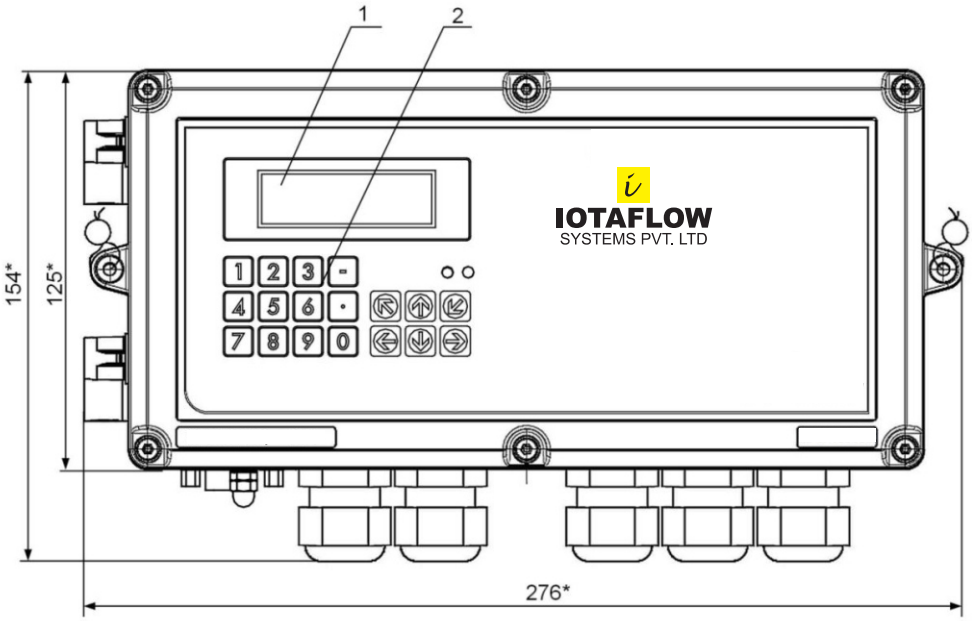
Sensing the fluid flow can be done with one, two or four beams (Fig. 3).



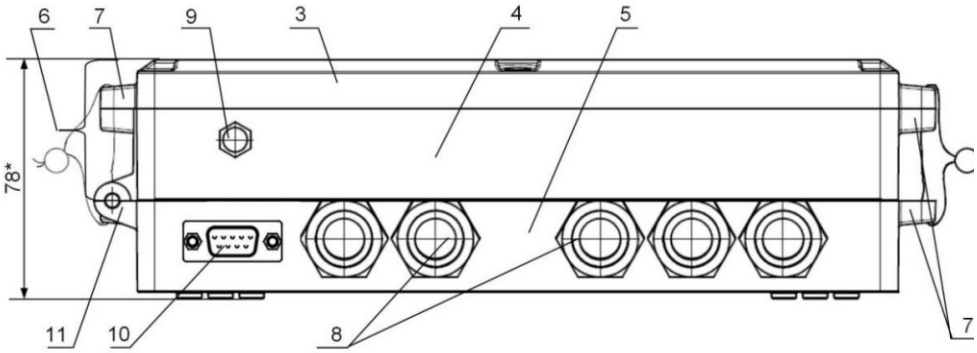
a) Single-beam diameter probing;
b) Two-beam sensing along the chord (for embedded PEA);
c) Double-beam diameter probing; d) four-beam sensing along mid-radius chords
(for embedded PEA, four sensors in 2 different sections of the pipeline)

View of the flowmeter's diagram

a) Front view

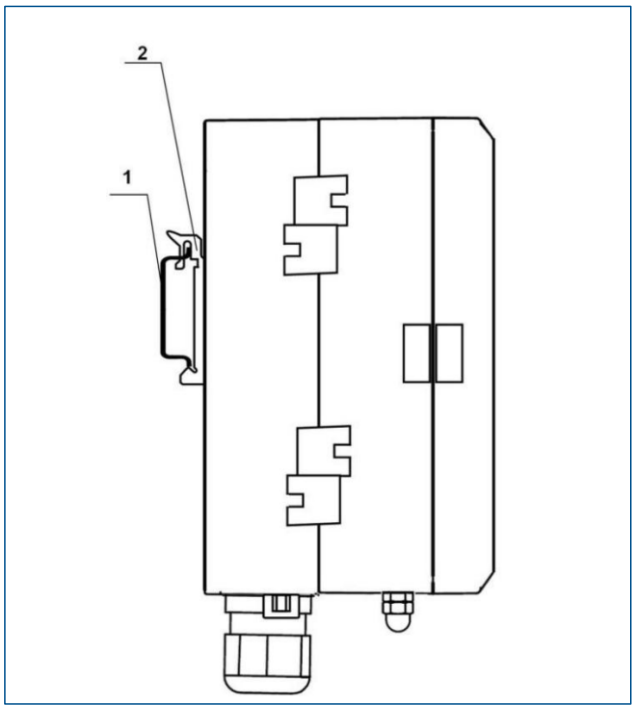


b) Botom view

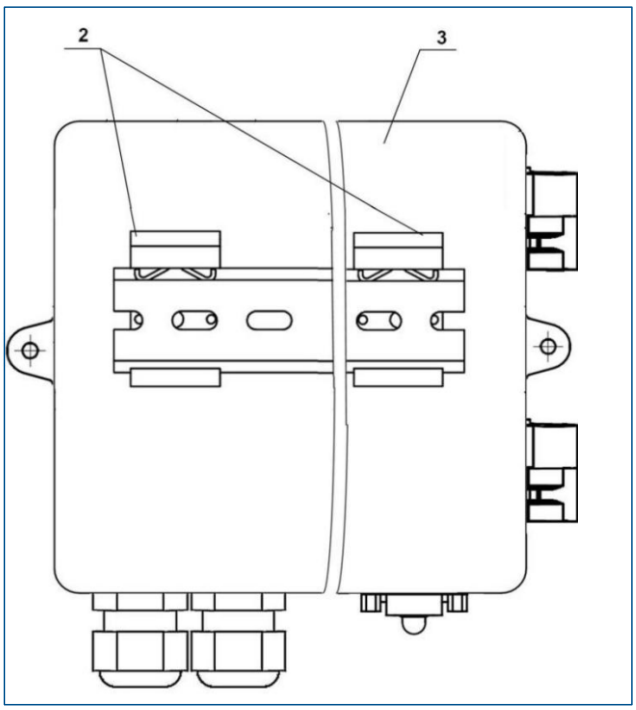


Reference size
1 Display:
2 Keyboard;
3 Converter and keyboard unit;
4 VIP block;
5 Switching unit;
6 Data processing subblock;
7 Eyes for installing service seals;
8 Pressure seals;
9 Ground terminal;
10 RS-232 interface connector;
11 Hinges on which the data processing subunit is folded.

c) Side view



d) Rear view



1 DIN rail
2 Mounting plates DRB01
3 VP.

Contact Us
IOTAFLOW SYSTEMS PVT. LTD.

C-2/13, Mayapuri Industrial Area Phase-II,
New Delhi - 110064 (INDIA)

+91-9910693866/ +91-7982674054

contact@iotaflow.com

Specifications subject to change without prior notice. © Iotaflow Systems Pvt. Ltd.